DiMI FY2000-2001 Proposal

Proposal ID: dimi00-10093

Title: Place-based eBusiness: Using interactive 3D virtual environments in support of web-based commerce

Research Field: Business Management

Campus/Lab: UC Berkeley

Status: New

Period: September 7, 2001 - September 6, 2004

Private Sponsor: LinuxLine, Inc.
San Jose, CA 95131

Participants

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Abstracts

ABSTRACTS: These descriptions are meant to serve as succinct, accurate descriptions of the proposed work when separated from the application.

TECHNICAL ABSTRACT FOR EXPERT REVIEWERS:

Abstract

The proposed research will explore the reciprocal relationships between architectural design and shopping in the context of e-business. Specifically, this grant will explore the impact of developing virtual archi-tectonic places to accommodate customized e-commerce: web 'places' that can support mass-customization of shopping for large products that must fit within the customers' existing contexts, such as their homes and offices. The product of the proposed research can be considered a configurator: a web-based environment and collection of tools that will support the procurement of a wide variety of products that need to be fitted together in some given context. The proposed environment will provide 3D visual means to explore and understand the containing environment (e.g., a room, an office, etc.), and selecting objects from product catalogs (also in 3D), along with the business tools needed to procure them and the personalized advise that supports the selection and fitting together of such products.

The proposed configurator will not be specific to one industry or market segment. Rather, it will be a prototype that can be adapted to any industry where standardized products, such as furnishings or appli-ances, must be fitted together in a given context. As such, with suitable adaptation, the configurator can be used to procure furniture for a house or an office, seats for an aircraft or train, and many other so-called 'situated' circumstances.

The research will test the hypothesis that architectonic place-making principles—such as the presence of other people, contextual frameworks, authenticity, adaptability, and multi-media tools (sound, color, texture), can enhance the quality of the shopping experience, thereby affecting a wide range of variables related to sales, such as volume, price, speed, level of customer satisfaction and general effectiveness of Internet or mail-order commercial activities. We will measure to what degree simulating the experiences gained from the physical world, or carefully controlling them in template based, personalized user inter-face, impacts the commercialization goals of shopping in the more automated Cyber-world. Specifically, the research and development activities under this grant will attempt to find out whether it makes eco-nomic sense to develop place-like web-based shopping experiences, instead of (or in addition to) the printed page, catalog-like experiences commonly employed today.

It is our objective to license various application components as they are developed in relation to this on-going program.

NON-CONFIDENTIAL, NON-TECHNICAL ABSTRACT FOR PUBLIC INFORMATION OR PROGRAM PROMOTION: State the application's broad, long-term objcetives and specific aims, making reference to the potential public benefits of the project relevant to California. Do not include proprietary or confidential information. This may be distributed before the funding descision has been finalized.

See above

PROJECT OBJECTIVE: Provide a one to two sentence, non-technical description of the project's overall objectives and key milestones.

See above

Sponsor Budget
First Year Budget Period

NON-CASH/IN-KIND CONTRIBUTION ONLY: Itemize each budget item by category (e.g. equipment, software, supplies, etc.) and by unit cost. Justify on Form G (Budget Justification).

LinuxLine, Inc.

In-Kind Contribution

Category	**Description** (make, model, or other description)	**Cost** (List Price or Other Cost)	**Modified Value** (List Price / 2)	**Total**
Supplies		0.00	0.00	
Equipment		0.00	0.00	
Other	System Documentation $6,000 Technical Assistance $60,000	66,000.00	33,000.00	**33,000.00**

Cash Contribution

Direct Costs		167,992.00	
Indirect Costs		67,840.16	**235,832.16**

Total LinuxLine, Inc. Year 1 Contribution **268,832.16**

DETAILED BUDGET FOR FIRST YEAR BUDGET PERIOD
DIRECT COSTS ONLY

FROM **9/7/2001** THROUGH **9/6/2002**

PERSONNEL (Applicant organization only)

NAME	ROLE ON PROJECT	APPOINTMENT (Months)	(%)	% EFFORT ON PROJECT	INSTITU-TIONAL BASE SALARY	DOLLAR AMOUNT REQUESTED SALARY REQUESTED	BENEFITS REQUESTED	TOTALS
TBA TBA	Postdoctoral Trainee	9	100	100	45,000.00	45,000.00	0.00	45,000.00
TBA TBA	Graduate Student	9	50	50	23,664.00	11,832.00	15,445.00	27,277.00
TBA TBA	Graduate Student	9	50	50	23,664.00	11,832.00	4,551.00	16,383.00
TBA TBA	Postdoctoral Trainee	9	100	100	45,000.00	45,000.00	0.00	45,000.00
TBA TBA	Graduate Student	3	100	100	7,423.00	7,423.00	668.00	8,091.00
TBA TBA	Graduate Student	3	100	100	7,423.00	7,423.00	668.00	8,091.00
Accounting Assistant TBA	Staff	12	100	25	37,800.00	9,450.00	1,890.00	11,340.00
Arie Segev	PI	12	100	38	87,800.00	33,364.00	4,828.00	38,192.00
Yehuda Kalay	Co-PI	3	100	100	34,712.00	34,712.00	3,471.00	38,183.00
TBA TBA	Graduate Student	9	50	50	23,664.00	11,832.00	15,445.00	27,277.00
Emily Ladner	Staff	12	100	30	37,800.00	11,340.00	2,268.00	13,608.00
				SUBTOTALS		229,208.00	49,234.00	278,442.00

CONSULTANT COSTS

0.00

SUPPLIES (Itemize by category)

Telephone/fax $3,042
Books $1,690
Software $6,610
Office and Computing Supplies $5,000
Research Materials $3,200

In-Kind 0.00
Cash 19,542.00 — 19,542.00

EQUIPMENT (Itemize)

2 Laptops $6,000
Portable Hi-res Computer Projector $4,000
DV Cam $3,000
Workgroup Color Laser Printer $2,000
3 Graphics Workstations $15,000

In-Kind 0.00
Cash 30,000.00 — 30,000.00

TRAVEL

Domestic Research Conf 2 PI $7,000
Bay Area Travel PI & Researchers $1,000

8,000.00

OTHER EXPENSES (Itemize by category)

In-Kind 33,000.00
Cash 0.00 — 33,000.00

TOTAL DIRECT COSTS FOR FIRST YEAR BUDGET PERIOD **368,984.00**

TOTAL MODIFIED DIRECT COSTS FOR FIRST YEAR BUDGET PERIOD
(Excluding costs that are not subject to indirect costs) **269.207.00**

Sponsor Budget

Second Year Budget Period

NON-CASH/IN-KIND CONTRIBUTION ONLY: Itemize each budget item by category (e.g. equipment, software, supplies, etc.) and by unit cost. Justify on Form G (Budget Justification).

LinuxLine, Inc.

In-Kind Contributio

Category	**Description** (make, model, or other description)	**Cost** (List Price or Other Cost)	**Modified Value** (List Price / 2)	**Total**
Supplies		0.00	0.00	
Equipment		0.00	0.00	
Other	System Documentation $6,000 Technical Assistance $60,000	66,000.00	33,000.00	**33,000.00**
Cash Contribution				
	Direct Cost		173,934.00	
	Indirect Cost		74,603.01	**248,537.01**
Total LinuxLine, Inc. Year 2 Contribution				**281,537.01**

DETAILED BUDGET FOR SECOND YEAR BUDGET PERIOD						FROM 9/7/2002	THROUGH 9/6/2003	
PERSONNEL (Applicant organization only) NAME	ROLE ON PROJECT	APPOINTMENT (Months)	(%)	% EFFORT ON PROJECT	INSTITU-TIONAL BASE SALARY	DOLLAR AMOUNT REQUESTED SALARY REQUESTED	BENEFITS REQUESTED	TOTALS
Arie Segev	PI	12	100	38	90,000.00	34,200.00	4,764.00	38,964.00
Yehuda Kalay	Co-PI	3	100	100	35,407.00	35,407.00	3,541.00	38,948.00
TBA TBA	Postdoctoral Trainee	9	100	100	46,000.00	46,000.00	0.00	46,000.00
TBA TBA	Graduate Student	9	50	50	24,138.00	12,069.00	15,754.00	27,823.00
TBA TBA	Postdoctoral Trainee	9	100	100	46,000.00	46,000.00	0.00	46,000.00
TBA TBA	Graduate Student	3	100	100	7,572.00	7,572.00	682.00	8,254.00
TBA TBA	Graduate Student	9	50	50	24,138.00	12,069.00	4,642.00	16,711.00
TBA TBA	Graduate Student	3	100	100	7,572.00	7,572.00	682.00	8,254.00
Accounting Assistant TBA	Staff	12	100	25	38,556.00	9,639.00	1,928.00	11,567.00
Administrative Assistant TBA	Staff	12	100	30	40,823.00	12,246.90	2,449.00	14,695.90
				SUBTOTALS		222,774.90	34,442.00	257,216.90
CONSULTANT COSTS								0.00
SUPPLIES (Itemize by category) Telephone/fax $3,062 Books $2,700 Software $5,000 Printing & copying $3,900						In-Kind Cash	0.00 14,662.00	14,662.00
EQUIPMENT (Itemize) Video & Graphics Hardware $15,000 Diagnostic Equipment $15,000						In-Kind Cash	0.00 30,000.00	30,000.00
TRAVEL Case Studies $2,500 Focus Group $6,000 Remote Monitoring & Beta Site $3,750								12,250.00
OTHER EXPENSES (Itemize by category) Case Studies $4,500 Programming Assistance $20,000 Design Assistance $8,800						In-Kind Cash	33,000.00 33,300.00	66,300.00
TOTAL DIRECT COSTS FOR SECOND YEAR BUDGET PERIOD								**380,428.90**
TOTAL MODIFIED DIRECT COSTS FOR SECOND YEAR BUDGET PERIOD (Excluding costs that are not subject to indirect costs)								**295,670.00**

Sponsor Budget

Third Year Budget Period

NON-CASH/IN-KIND CONTRIBUTION ONLY: Itemize each budget item by category (e.g. equipment, software, supplies, etc.) and by unit cost. Justify on Form G (Budget Justification).

LinuxLine, Inc.

In Kind Contribution

Category	**Description** (make, model, or other description)	**Cost** (List Price or Other Cost)	**Modified Value** (List Price / 2)	**Total**
Supplies		0.00	0.00	
Equipment		0.00	0.00	
Other	System Documentation $6,000 Technical Assistance $60,000	66,000.00	33,000.00	**33,000.00**
Cash Contribution				
	Direct Cost		178,393.00	
	Indirect Cost		84,151.77	**262,544.77**
Total LinuxLine, Inc. Year 3 Contribution				**295,544.77**

DETAILED BUDGET FOR THIRD YEAR BUDGET PERIOD
DIRECT COSTS ONLY

FROM **9/7/2003** THROUGH **9/6/2004**

PERSONNEL (Applicant organization only)

NAME	ROLE ON PROJECT	APPOINTMENT (Months)	(%)	% EFFORT ON PROJECT	INSTITUTIONAL BASE SALARY	DOLLAR AMOUNT REQUESTED SALARY REQUESTED	BENEFITS REQUESTED	TOTALS
Arie Segev	PI	12	100	38	92,347.00	35,091.86	4,813.00	39,904.86
Yehuda Kalay	Co-PI	3	100	100	36,115.00	36,115.00	3,612.00	39,727.00
Administrative Assistant TBA	Staff	12	100	30	44,090.00	13,227.00	2,645.00	15,872.00
TBA TBA	Postdoctoral Trainee	9	100	100	47,000.00	47,000.00	0.00	47,000.00
TBA TBA	Graduate Student	9	50	50	24,620.00	12,310.00	16,169.00	28,479.00
TBA TBA	Graduate Student	9	50	50	26,454.00	13,227.00	5,290.00	18,517.00
TBA TBA	Graduate Student	3	100	100	7,423.00	7,423.00	695.00	8,118.00
TBA TBA	Graduate Student	3	100	100	7,423.00	7,423.00	695.00	8,118.00
TBA Accounting Assistant TBA	Staff	12	100	25	40,483.00	10,120.75	1,721.00	11,841.75
				SUBTOTALS		181,937.61	35,640.00	217,577.61

CONSULTANT COSTS

Programming Consultant $50,000
Application, Design and Development $60,000

110,000.00

SUPPLIES (Itemize by category)

Copying $4,000
Document & Books $1,700

In-Kind	0.00	
Cash	5,700.00	5,700.00

EQUIPMENT (Itemize)

In-Kind	0.00	
Cash	0.00	0.00

TRAVEL

Travel to Conferences $12,000
Bay Area Travel PI & Researchers $1,000

13,000.00

OTHER EXPENSES (Itemize by category)

Printing & Publication $5,500
Mailing $2,000
Telephone/fax $3,083

In-Kind	33,000.00	
Cash	10,483.00	43,483.00

TOTAL DIRECT COSTS FOR THIRD YEAR BUDGET PERIOD **389,760.61**

TOTAL MODIFIED DIRECT COSTS FOR THIRD YEAR BUDGET PERIOD
(Excluding costs that are not subject to indirect costs) **333,911.85**

Budget for Entire Proposed Project Period

A. Project Summary

Budget Category		Year1		Year2		Year3		Total Project
Salaries and Benefits			278,442.00		257,216.90		217,577.61	**753,236.51**
Consultant Costs			0.00		0.00		110,000.00	**110,000.00**
Supplies	In-kind	0.00		0.00		0.00		
	Cash	19,542.00	19,542.00	14,662.00	14,662.00	5,700.00	5,700.00	**39,904.00**
Equipment	In-kind	0.00		0.00		0.00		
	Cash	30,000.00	30,000.00	30,000.00	30,000.00	0.00	0.00	**60,000.00**
Travel			8,000.00		12,250.00		13,000.00	**33,250.00**
Other Expenses	In-kind	33,000.00		33,000.00		33,000.00		
	Cash	0.00	33,000.00	33,300.00	66,300.00	10,483.00	43,483.00	**142,783.00**
Total			**368,984.00**		**380,428.90**		**389,760.61**	**1,139,173.51**

B. Direct Costs Summary

	DiMI	Private Sponsor		
	Cash	Cash	Modified In-Kind	Total
Year 1 Direct Costs	167,992.00	167,992.00	33,000.00	368,984.00
Year 2 Direct Costs	173,494.90	173,934.00	33,000.00	380,428.90
Year 3 Direct Costs	178,367.61	178,393.00	33,000.00	389,760.61
Total	**519,854.51**	**520,319.00**	**99,000.00**	**1,139,173.51**

C. Maximum Allowable In-Kind Match

	DiMI Total Direct Costs (from Section B)		In-Kind Ceiling		Maximum Allowable Modified In-Kind for Match	Total Modified Inkind (from Section B)	Additional In-Kind (Total Modified Less Maximum Allowable)
Private Sponsor	519,854.51	x	20%	=	103,970.90	99,000.00	0.00

D. Indirect Costs Summary

	Year 1	Year 2	Year 3
1. Modified Direct Costs	269,207.00	295,670.00	333,911.85
2. Indirect Cost Rate	50.40%	50.40%	50.40%
3. Private Sponsor Cash-to-Total Cash Ratio	50.00%	50.06%	50.00%
4. Indirect Costs Paid by Sponsor (D1 x D2 x D3)	**67,840.16**	**74,603.01**	**84,151.77**
Total Indirect Costs Paid by Sponsor (Years 1+2+3)			**226,594.95**

E. Contributions Summary

			DiMI	Private Sponsor	
Direct Costs:	Cash		519,854.51		520,319.00
	In-Kind:	Match Amount	-0-	99,000.00	
		Additional Amount	-0-	0.00	99,000.00
Indirect Costs			-0-		226,594.95
Voluntary Contribution			-0-		0.00
Total			**519,854.51**		**845,913.95**

Budget Justification

BUDGET JUSTIFICATION

All salaries are based on current rates as of 10/01/00. Faculty and Graduate Student Researchers are projected at 2% annually. Clerical staff increases are projected at 8% annually. Salary rates of GSRs are as follows:

GSR Salary Rate: $14.22 per hour
Academic Year 50% effort
Summer 100% effort

Benefits rates are based on current benefits being paid with 2% estimated increase in years 2 & 3.

GSR Fees are included in Benefits with 2% increase in years 2 & 3.
Student Health Insurance Plan 108.54 per semester
Fee Remission 2157.25 "
Non-Resident Tuition 7604.25 "

Equipment:
Year 1 2 Laptops
Portable Hi-Res Computer Projector
DV Cam
Workgroup Color Laser Printer
3 Graphics Workstations
Year 2 Video and Graphics Hardware
Diagnostic Equipment

Travel
Year 1: The budget includes 2 conference trips and Bay Area travel
Year 2: There are funds budgeted for travel for Case Studies, Focus Groups and Remote Monitoring and Beta Site support.
Year 3: Funds are budgeted for travel to conferences and the Bay Area for the purpose of dissemination.
Conference expenses include airfare, per diem, registration fees and ground transportation.

Other Direct Costs:
a. General office supplies needed to set up and maintain project research.
b. Research materials and books.
c. Telephone/fax and mailing costs.
d. Software
e. Printing and copying for research and dissemination.
f. Publication and mailing of research documents resulting from this project.
g. Consultants hired to advise and assist with technical, programming and design.

Indirect costs are charged at the current industry rate of 50.4% of the modified direct costs of the industry sponsor`s portion of the award. DIMI funds are exempt from Indirect Costs as are equipment, GSR tuition, fees and SHIP costs.

Other Support - Current

None

Other Support - Pending

None

Resources

FACILITIES: Specify the facilities to be used for the conduct of the proposed research. Indicate the performance sites and describe capacities, pertinent capabilities, relative proximity, and extent of availability to the project. Under "Other," identify support services such as machine shop, electronics shop, and specify the extent to which they will be available to the project.

Laboratory:

UC Berkeley has offered space in the Marchant Building to house the equipment to be used for this project.

Clinical:

None

Animal:

None

Computer:

All academics, staff and GSRs are equipped with their own PCs. Printers are available in workstations or offices and in the Haas Computer Resource Center. The Haas School of Business at UC Berkeley has full time staff in electronics who are available for assistance both with the hardware and software of computers.

Office:

Office space is provided for the PI and CO-PI who are both full professors at UC Berkeley. IMIO has workstations in Room F402 available for the other administrative and accounting staff. In addition, there are rooms in the Haas buildings available to the graduate student researchers on this project.

Other:

None

MAJOR EQUIPMENT: List the most important equipment items already available for this project, noting the location and pertinent capabilities of each.

None

Private Sponsor Data

Private Sponsor: LinuxLine, Inc.
780 Montague Exwy., Suite 306
San Jose, CA 95131
Phone: 408-435-8810
Email: John@aqualux.net; info@datarise.com

Nature Of California Business: Software Engineering Firm

Year Founded: 2001

California Employees: 8

US Employees: 8

Worldwide Employees: 35

Employees in R&D: 32

Benefit to Company: The proposed research program will benefit LinuxLine, Inc., located in California, by providing the company with cutting edge, core technology and foundational software intended for further development and adoption of sophisticated E-Commerce selling and buying side systems at highly attractive pricing. The base technology research focuses on generating a new level of collaborative selling environments that allow many new players, faster and easier customization of combined sales and services, in addition to building on sales volumes whilte enhancing customer service and satisfaction levels. All of these end-user benefits build synergy withthe engineering skills of LinuxLine, Inc., the company`s planned products and services, and its major business development and partnering plans to provide the company with new growth potential. The company plans to actively build commensurate products, services and partners for this intelligent framework research that is well suited to the open-source Linux operating system and associated environments. While our focus is on building business in California, we further intend to actively pursue a variety of attractive vertical selling markets that would extend the reach of our business for maximum E-Commerce adoption of program results.

Private Sponsor Data

Benefit to California: Building core technology and sophisticated software, networking and other E-Commerce related systems, products and services on the open-source and free-of-charge, Linux operating system kernel helps guarantee rapid industry adoption, further private and public investments and extended partnering solution efforts. Any company may use and / or disseminate open-source Linux-based products also free of charge. The highly cost effective nature of Linux enhances the ability to provide further technological sophistication to products, and still offer the products and services supporting them at levels affordable to small and medium sized businesses. This process provides a variety of vertical market industries with significant competitive advantages, which result in increased sales and job opportunities specifically for the benefit of California businesses. Again, the extended results of these added opportunities for California business further translate into advanced reach into global market economies, maintaining and increasing California`s business leadership position.

ARIE SEGEV



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Curriculum Vitae (in PDF format)

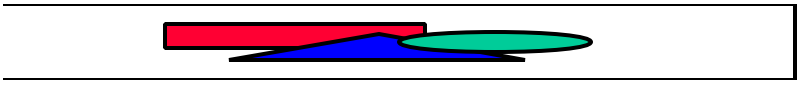
BS, industrial engineering and management, Technion, Israel Institute of Technology
MS, industrial engineering and management, Technion, Israel Institute of Technology
MS, operations research, University of Rochester
Ph.D., computers and information systems, University of Rochester

At Haas since 1983

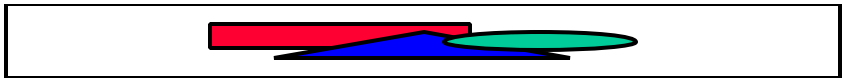
1983-present Professor, Haas School of Business
1999-present Fisher Center for Information Technology & Marketplace Transformation, Haas School of Business
1994-99 Director, Fisher Center for Management & Information Technology Management Program
1989-94 Director, Information Technology Management Program, Center for Research in Management, Haas School of Business
1987-present Faculty research, Information and Computing Sciences Division, Lawrence Berkeley National Laboratory

- Editorial board member: *IEEE Transactions on Data & Knowledge Engineering, Information Systems Research, INFORMS Journal on Computing, Information Technology and Management, Electronic Markets*
- Committee member: program and organization committees for numerous academic and industry conferences
- Consultant: The A.T. Kearney, Concours Group, IBM, and Information Week
- Business-to-business Internet procurement and supply chain

- Negotiations and brokering in E-commerce
- Information management
- Data warehousing and mining
- Web-based information systems design
- "A Bigger Picture of Internet Security -- The Case of Bank of America." (with J. Porra and M. Roldan). *Communications of the ACM*. 41 no.10, (1998): 81-87.
- "A Brokerage Framework for Internet Commerce." (with M. Bichler). *Journal of Parallel and Distributed Databases*. To Appear. (1999).
- "An Electronic Broker for Business-to-Business Electronic Commerce." (with M. Bichler and C. Beam). *International Journal of Cooperative Information Systems*. To Appear. (1999).
- "Transforming Business in the Marketspace: Strategic Marketing and Customer Relationships." (with S. Dutta & S. Kwan). *Communications of the ACM*. To Appear. (1999).
- "Procurement in the Internet Age - Current Practices and Emerging Trends (Results From a Field Study)." (with J. Gebuaer & C. Beam). Fisher Center Working Paper 98-WP-1033, University of California, Berkeley, 1998.
- "Internet-Based EDI Strategy." (with J. Porra & M. Roldan), *Decision Support Systems*. 21 no. 3 (November 1997): 157-170.



- BA 208, Information Technology Management, Spring 99
- BA 296, Cases in Electronic Commerce, Spring 99
- BA 296, Strategic Information Management, Spring 99



- IBM Fellowship (Advanced Commerce Institute), 1998-1999
- Fisher Center Grants, 1994-1999
- Lawrence Berkeley National Laboratory Grants, 1987-1997
- National Science Foundation Grants, 1988-1995
- Best-Paper Award, ACM Special Interest Group on Management of Data, 1987

YEHUDA E. KALAY

Professor Kalay received his B.Arch. and M.Sc. degrees from the Technion, Israel Institute of Technology, in 1976 and 1978, respectively, and earned a Ph.D. from Carnegie Mellon University (Pittsburgh, PA) in 1982.

He was appointed Professor of Architecture in the College of Environmental Design at the University of California, Berkeley, in 1992. Prior to his tenure at Berkeley, for 10 years Professor Kalay was a Professor of Architecture and Adjunct Professor of Computer Science of the State University of New York at Buffalo, where he also served as director for research of the Department or Architecture. In addition to teaching at Carnegie Mellon University, SUNY Buffalo and UC Berkeley, Professor Kalay has taught at the Technion (Israel), the University of Rome (Italy), and at Stanford University. He was awarded two times the Lady Davis Visiting Professorship at the Technion, in 1988 and 1997.

Professor Kalay is a founding member and past president of ACADIA (Association for Computer Aided Design In Architecture). He is Editor-in-Chief of *Automation in Construction*, an international refereed journal published by Elsevier Science in Oxford, UK and has been keynote speaker in numerous conferences all over the world.

Kalay's research focuses on performance-based design, collaboration, and extending the principles of architectural design to Cyberspace. It include geometric and semantic modeling, knowledge representation, performance evaluation, collaborative design, situated e-commerce, and other related topics. Over the past 20 years, Professor Kalay's research has been sponsored by the National Science Foundation, the National Endowment for the Arts, the U.S. Army Corps of Engineers, the Italian National Council for Research (CNR), the Israeli Ministry of Housing, the Universities of California and New York, as well as many corporate grants, including: Boeing, Sun Microsystems, Sigma Design, The Maedl Group, Intel, CARTESIANA (Italy), and Intelligence Artificielle (Switzerland). The results of these research projects have been published in over 100 scholarly papers.

Professor Kalay has published five books, including: "Computability of Design" (Wiley, 1987), "Modeling Objects and Environments" (Wiley, 1989), "Evaluating and Predicting design Performance" (Wiley, 1992), and "Knowledge-based Computer-Aided Architectural Design" (Elsevier, 1994).

Place-based eBusiness:
Using interactive 3D virtual environments in support of web-based commerce

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Aims

The proposed research will explore the reciprocal relationships between architectural design and shopping in the context of e-business. Specifically, this grant will explore the impact of developing virtual architectonic places to accommodate customized e-commerce: web 'places' that can support mass-customization of shopping for large products that must fit within the customers' existing contexts, such as their homes and offices. The product of the proposed research can be considered a *configurator*: a web-based environment and collection of tools that will support the procurement of a wide variety of products that need to be fitted together in some given context. The proposed environment will provide 3D visual means to explore and understand the containing environment (e.g., a room, an office, etc.), and selecting objects from product catalogs (also in 3D), along with the business tools needed to procure them and the personalized advise that supports the selection and fitting together of such products.

It is the nature of new technologies to emulate existing ones when they are first introduced. Examples include the first automobiles, which were likened to 'horseless carriages,' and early movies, which were mere celluloid records of stage-plays. E-commerce has adopted the mail-order catalog metaphor: a shopping mechanism that was invented by Montgomery Ward some 150 years ago with the intention of reaching farmers who were unable to travel to physical stores. In many respects, e-commerce has merely replaced the paper catalog with an on-line one, and added the data management capabilities that ensue from computer-based, rather than paper-based, inventory management.

But mail-order shopping was intended to be a second-best alternative, not a substitute for in-store shopping. That is why most retailers who have, in the past 50 years, launched their own forays into some form of mail order business, still find it necessary to establish and/or maintain brick and mortar stores, where products can be viewed and touched, and were shoppers can mingle and be influenced by each other, thereby creating the all-important 'shopping mood' or socialization experience (especially around holidays, like Christmas). The need for fully sensorized, physically and socially interactive shopping is so highly valued that even the enormous savings associated with mail-order vs. physical stores have not been able to sway retailers to abandon the expensive, human resource intensive and liability laden brick & mortar model, in favor of the catalog only model. Our hypothesis further suggests that traditional catalogs, even those employing three dimensional, rotatable photographic models of the products, do not adequately satisfy customer demands for experiencing or sampling products prior to a purchase commitment.

The need for a fully sensorized, place-specific shopping experience is strongest where the newly purchased products must fit in with an existing context, comprising both objects and spaces, while adding some purpose or value without interfering or diminishing the value of the established set-up. For example, a new sofa or entertainment center must fit in within the context of a given living room. A new refrigerator must fit space requirements, power and water hook-ups, and existing kitchen appliances. And a new desk must fit within the context of existing office space, style, furniture, lighting fixtures, and even the habits and attitudes of office workers. This is why most furniture showrooms find it necessary to set up faux living rooms, kitchens and offices, in order to provide shoppers with a feel for the relationships between the items they seek and other furniture, or to simulate the space into which the new items will fit when they are placed in the shopper's own home or office environment. Stores like IKEA, Ethan Allen, SamClar, and others devote literally acres of floor space to displaying their wares 'in context.' Still, buyers frequently express distress at their inability to experience various sofas or desks in their own living or work places, where they can be viewed in combination with items they already own. The buyer's inability to envision how the store-displayed items will fit in his/her own home or office often results in an entirely different experience than intended, and contributes to returns and slow inventory turns. There is literally no substitute for 'in home' or 'in office'

personalized shopping. Yet, the size and transportation costs of such big ticket items as sofas, entertainment centers and office furniture have made in-home or in-office experiences virtually impossible, until now.

It is our contention that, as the Web matures and assumes more fully its role as an information *space* rather than as a communication *network*, it will be possible to combine the advantages of in-shop retail with mail-order savings, and to avail to shoppers the contextualized, in-home/office experience they crave when shopping big ticket items.

Specifically, we contend that the Internet has the potential to streamline the process of procuring such items, by combining the benefits of physical (or rather, virtual) contextuality with the business practices and coordination methods of e-commerce. It should be possible, for example, to build a model of a customer's office, and see how desired furnishings can be placed in it. How light, streaming in from existing windows, support—or interfere—with the activities of the office; where lighting fixtures may need to be added (or removed); where electrical and data outlets need to be installed, and so on. It should even be possible to test the proposed layout against building codes, simulate emergency egress, potential security problems, and people movements.

Moreover, furniture, appliances and other larger objects often require adjustments and modifications of the environment into which they go. Lighting fixtures, carpeting, window treatments, air conditioning, electrical, water and gas hook-ups are only a few examples of re-design and remodeling necessary to accommodate newly purchased items. Such changes often cannot be envisioned without situating the new items within the existing context, and without the specific expertise and assistance of design professionals and contractors. By making the model available for inspection by other trades, such as electricians, carpenters, etc., a comprehensive cost estimation can be derived, and alternative layouts can be explored in the context of on-line and personalized suggestive sales. Experts in space planning (architects, interior designers, security experts, etc.) can avail their services on-line, either in person (real time) or by the proxy of intelligent agents, to assist the shopper in making intelligent choices. This will make the decision making process better informed, leading to better design, higher productivity, and reduced costs.

The research will test the hypothesis that architectonic place-making principles—such as the presence of other people, contextual frameworks, authenticity, adaptability, and multi-media tools (sound, color, texture), can enhance the quality of the shopping experience, thereby affecting a wide range of variables related to sales, such as volume, price, speed, level of customer satisfaction and general effectiveness of Internet or mail-order commercial activities. We will measure to what degree simulating the experiences gained from the physical world, or carefully controlling them in template based, personalized user interface, impacts the commercialization goals of shopping in the more automated Cyber-world. Specifically, the research and development activities under this grant will attempt to find out whether it makes economic sense to develop place-like web-based shopping experiences, instead of (or in addition to) the printed page, catalog-like experiences commonly employed today.

It is our goal, through this project, to develop prototype tools and methods that will demonstrate the efficacy of the scenario that was described above, so it can be tested and applied to a wide range of businesses that require place-based, contextualized, e-commerce.

The proposed configurator will not be specific to one industry or market segment. Rather, it will be a prototype that can be adapted to any industry where standardized products, such as furnishings or appliances, must be fitted together in a given context. As such, with suitable adaptation, the configurator can be used to procure furniture for a house or an office, seats for an aircraft or train, and many other so-called 'situated' circumstances.

It is our objective to license various application components as they are developed in relation to this on-going program.

Relevance

E-commerce, and its attendant technologies have been the backbone of the New Economy, a product of California's Silicon Valley. It was promoted as a substitute for brick & mortar shopping, and touted for its point-and-click convenience. But the lofty aspirations of web-based shopping proponents were humbled by the recent slump in the technology sector of the economy overall, and the demise of the so-called 'dot-coms' in particular. We contend that one of the factors responsible for retarding the expected broad adoption of web-based shopping has been its almost exclusive reliance on the limited mail-order catalog metaphor, rather than on broader architectural metaphors.

There is no denying that web-shopping is convenient. But convenience, in-and-of itself, is not enough. The demise of Webvan.com and Peapod.com are a case in point: a brick & mortar supermarket cannot begin to compete with the convenience of the on-line grocery shopping model, which bundles wide variety and door-to-door delivery with competitive prices. Still, few people have used such services more than once. Conversely, the continuing success of brick & mortar stores like Barns & Noble, even in the face of fierce competition from web-only booksellers like Amazon.com, shows that shoppers crave for something more than mere convenience.

History has demonstrated that this added value is the place-based, socially-enhanced ***event*** of shopping.[1] People have been flocking to markets for centuries, not only to transact business, but for the 'event' itself. This 'event' comprises meeting old friends and making new ones, engaging in discussion, watching tournaments, entertainment, and even mischief. It is such 'events' that have fueled the evolution of markets into the civic centers of cities and villages, and into today's shopping malls and department stores.[2] In contrast, web-based shopping feels sterile: there may be millions of shoppers at any given time browsing Amazon.com's 'bookshelves,' but each one of them feels alone.

The activity of shopping is thus a social activity. Shoppers have a need for fully-sensorial engagement with the merchandise they are about to purchase, as well as with each other. Such needs grow with the uniqueness of the product, with its price, and, particularly, when the newly purchased product must fits in with existing possessions and environments.

It is our contention that by leveraging the convenience of the Web to support the psycho-social needs of shoppers through place-based shopping we can help to re-vitalize e-commerce, contribute to reversing the demise of the dot-coms, and open new possibilities for enterprising startups.

Our unique alliance of researchers from the disciplines of architecture and business promises to provide rigor to both the *setting* and the *activity* of shopping, something current efforts have neglected to do on one side or another.

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1 J.F. Sherry (1997). *Servicescapes: The Concept of Place in Contemporary Markets*, Lincolnwood, IL: NTC Business Books.

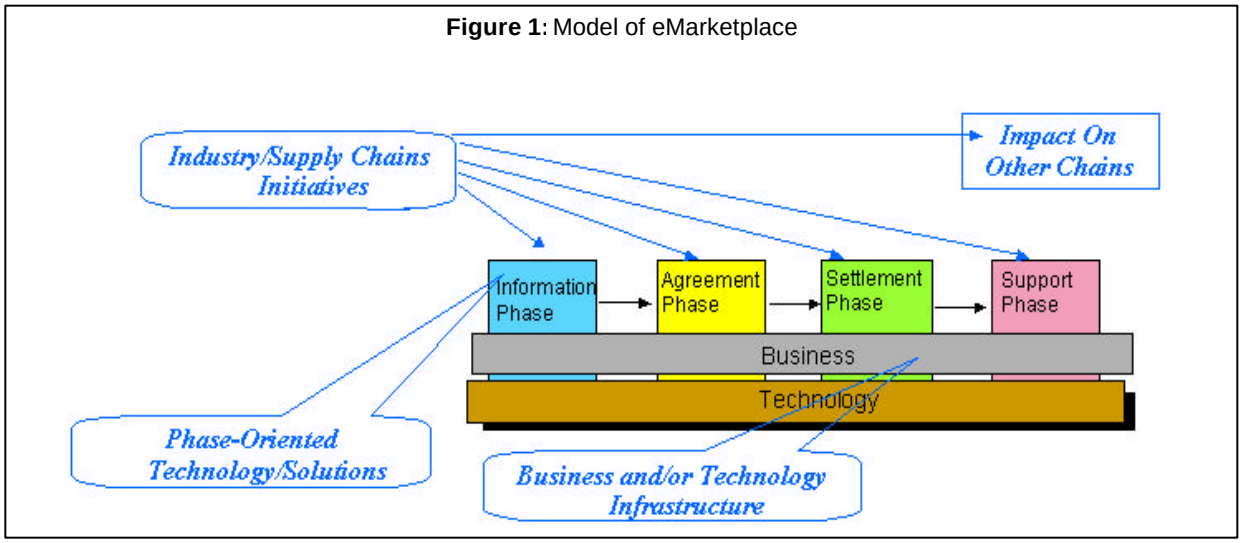
2 Kmart's recent re-introduction of the 'Blue Light Special' is, in part, a recognition of the importance of the shopping 'event,' as is Barns & Noble introduction of coffee shops in their bookstores.

Background

In recent years, an accelerated rate of Internet-related technology introductions has radically changed the business landscape of both small and large companies. Marketplace developments can be segmented as shown in Figure 1. A simple business transaction model assumes four phases (which can be interleaved):

- *Information Phase*, where requirements are determined, and products and their sourcing are identified. This phase can be very simple in cases where the buyer knows the exact product specifications, or very complex, involving internal and *Internet* catalog searches, structuring a Request for Proposal (RFP), or collaborative design.
- *Agreement Phase,* where negotiations take place. Again this phase can be very simple (or even absent from the transaction level, e.g., in the case of master buying contracts), or very complex with negotiations involving various attributes, such as price, quality, delivery dates, and payment terms. The negotiations may be 1:1, 1:M, or M:N and employ various negotiation techniques just as bidding, reverse auctions, etc.
- *Settlement Phase,* which includes the payment, delivery and any service work agreed upon. The complexity of the settlement phase is dependent on the number of parties involved in the execution of the contracted deal, and particular coordination problems at the seller and/or buyer's sites.
- *Support Phase,* which includes returns, warranty or non-warranty service, and variety of after-sale activity, which may or may not involve the original seller, such as property management. In very simple cases this phase may not be present.

Figure 1: Model of eMarketplace

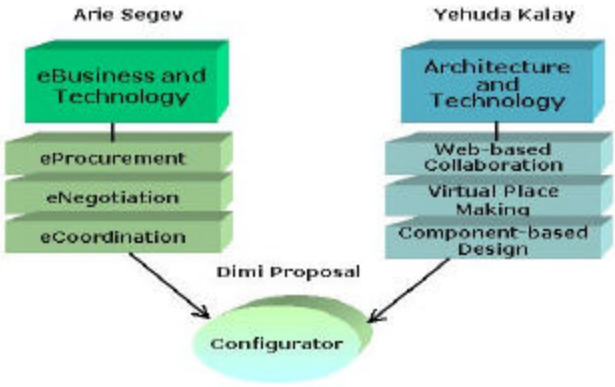


The marketplace activities include technology infrastructure such as communication, security certification and other tasks that require standardization or interoperability with various standards. The business infrastructure includes software and data components agreed upon through standardization activities (e.g. EDI), or through ad-hoc vendor initiated standards. Some of the "standards" are industry specific (e.g., RosettaNet in the IT supply-chain industry) or function/product specific (e.g., office supplies). In addition many technology solutions are targeted at specific phases, e.g. catalog search in the information phase, or particular payment system at the settlement phase. Some of the major problems that were contributing factors in the recent technology downturn were the lack of total business process solution, insufficient

functionality for specific tasks, and integration problems with legacy systems as well as more recent systems (both within the company and across trading partners).

The proposed research addresses the broader issues of eBusiness while at the same time it incorporates domain-specific functional and technology consideration. Moreover, it proposes innovative approaches to a problem, which is naturally interdisciplinary. The unique strength of this proposal is the combined expertise and experience of the project principals. As Figure 2 shows, a central part of the project is the design of a solution Configurator that draws on existing research of the two PIs. The proposed research draws upon the subjects listed, which represent research areas that have been pursued by the PIs in the last five years, frequently collaboratively with industry.

Figure 2: Research Domain Synergies



In the eBusiness and Technology area, CITM has been pursuing industry-oriented eBusiness research led by Professor Segev since 1996. A significant emphasize has been placed on B2B eProcurement, auctions and negotiations in eCommerce, and Information Management and Interoperability. In addition, recently started research on eCoordination is highly relevant to the proposed research. Figure 3 illustrates the nature of the problem addressed. That work is in the context of purchasing a (possibly customized) solution from multiple sources, and explores different protocols (both business processes and data interoperability). In the proposed research (see the Method section), addressing the coordination problem is highly important. Relevant activities and publications of Professor Segev in the eBusiness and Technology area can be found at:

http://haas.berkeley.edu/citm/forum, and at
http://haas.berkeley.edu/citm/research

In the architecture area, the proposed research draws upon three areas of design knowledge: (1) virtual place-making; (2) component-based design; and (3) Web-based collaboration. Additional publications in these areas can be found at:

http://arch.ced.berkeley.edu/people/faculty/kalay .

Method

The proposed research project will be done in the general context depicted in Figure 7. The project will be done from the Seller's perspective taking into consideration the Buyer's and suppliers' perspectives. The Seller (***S***) can be a retailer, wholesaler, manufacturer, or an intermediary. The Buyer (***B***) can be either a consumer or a business. It is assumed that ***B*** has already decided to explore a purchase of a solution ***L*** from ***S***, and the components of ***L*** can be assembled from ***S*** and from other sellers in groups ***G1*** and ***G2***. The solid arrows in the figure indicate the sellers ***B*** can buy from. Sellers in ***G1*** are those that ***S*** has a business relationship with (***G1*** can intersect ***G2***). ***G3*** represents the next level of suppliers in the supply chain. We assume under the scope of this project that suppliers in ***G3*** are inaccessible to ***B***, and may or may not be transparent to ***S***. The chosen context of office furnishings makes the results applicable to both consumer and business buying.

It should be emphasized that this project does not deal with buying standard office supplies, which is supported by various desktop procurement systems (DPS) and marketplaces. Rather, the emphasize here is on configuring a solution which involves customization; negotiation of functional requirements and variables such as price, delivery dates, and coordination. The various processes are distributed across multiple sites and multiple business entities. Since this project is done from the Seller's perspective, the primary system components will reside at a portal site that the Seller owns. In the case of consumer and small business buying, we expect the client functionality to be provided through the browser, and in the case of larger business buying, links to DPS and ERP systems might be needed. These more complex links are likely to be needed for the interactions between ***S*** and ***G1*** . While we assume that ***S*** does not have business relationships with ***G2***, there may be a benefit in ad-hoc linking to sellers in ***G2*** in order to provide ***B*** with a total solution. The scope of this project is limited to cases where ***G3***'s suppliers are transparent to ***B*** and ***S***. The specific division of the functional components among servers and clients, their physical locations and interoperability issues are a topic of the proposed research.

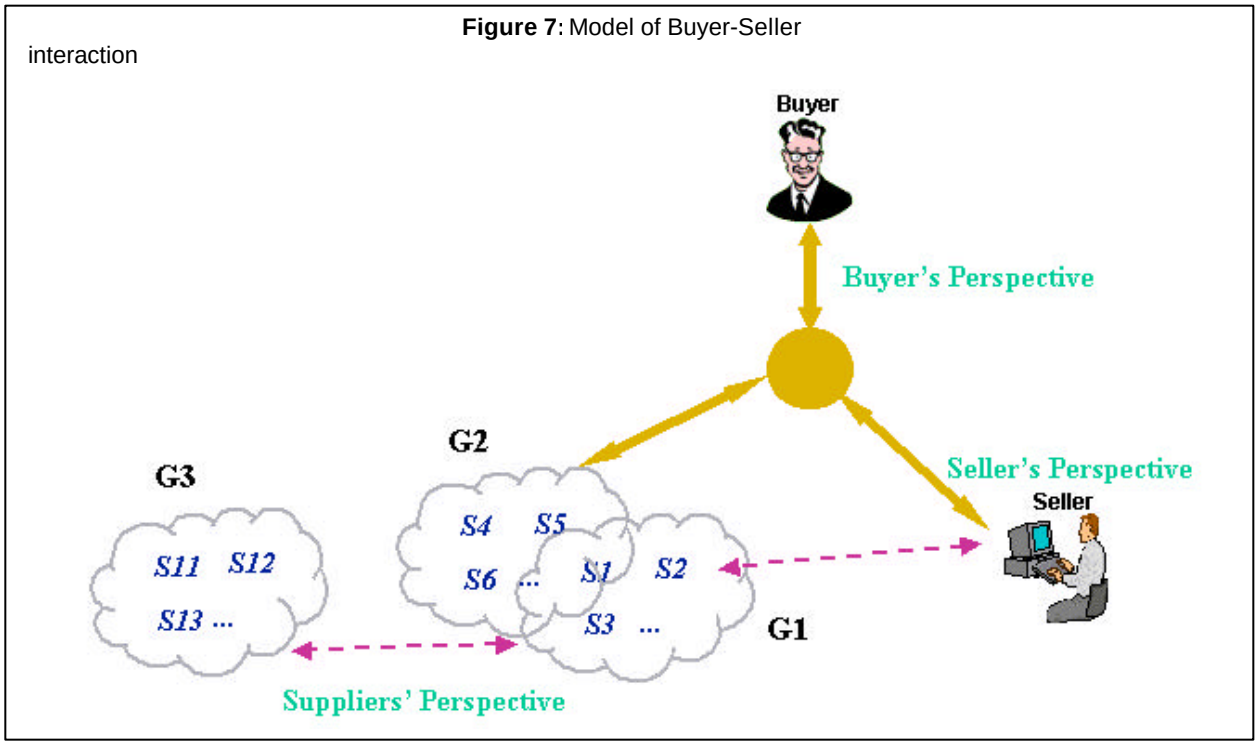
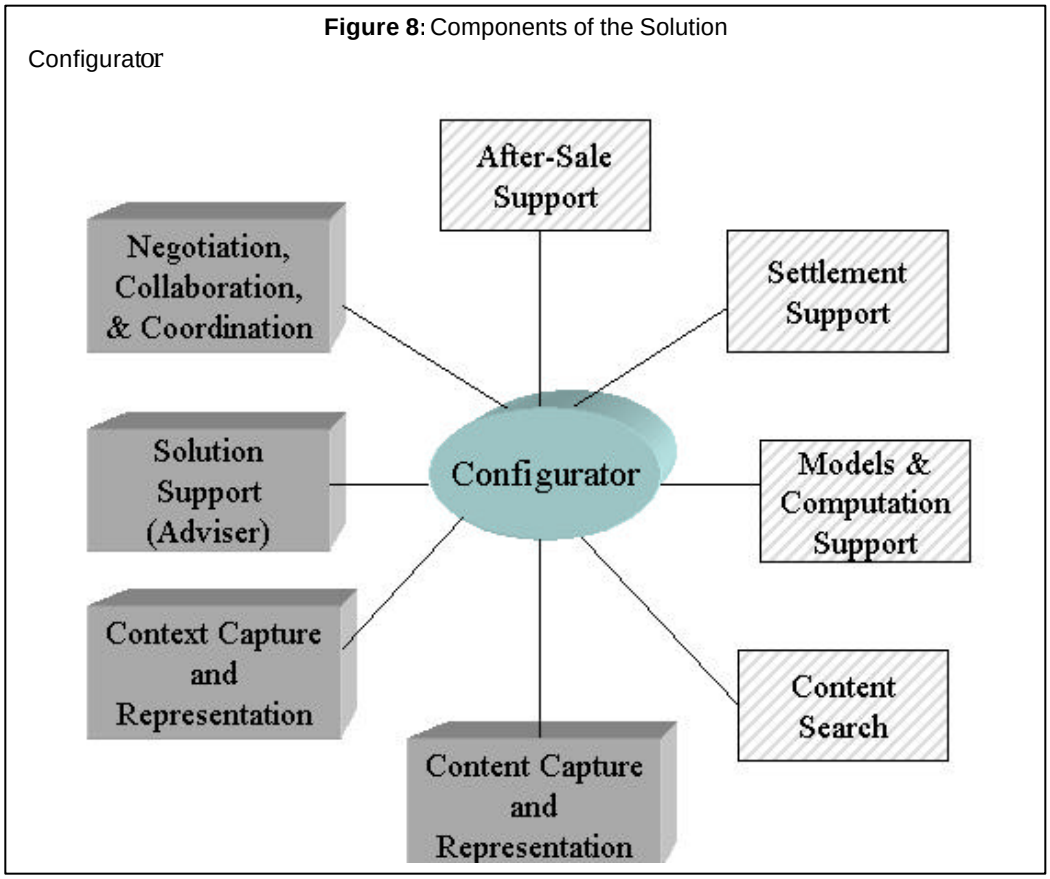


Figure 7: Model of Buyer-Seller interaction

Figure 8 depicts the functional components of the configurator; the 3-D blocks indicate those areas to be investigated in-depth throughout the project term. The striped blocks represent important functional components but their specific design is beyond the scope of this research. However, the configurator will be designed to hook-up with these other functional components. In the case of Content Search we expect to use existing software as part of the database system or an e-Catalog system (if one can be integrated with the primary functional components of the configurator). In addition to developing the components of the configurator interleaving the various functions in the configuration process is an important part of the proposed research.

Figure 8: Components of the Solution Configurator



To better manage the project, we have separated the work into major functional tasks. It is important to note that these tasks are interrelated and are presented here at a high level. The tasks are shown below followed by an overview of the plan for their implementation.

Task 0: Mock-up

Develop a comprehensive, interactive, story-board like mock-up that will be used to understand, explore, test, and communicate the ideas underlying the project. We have found mock-ups to be a useful tool to tease out ideas and test them before too much effort has been committed to following one development path or another. Like architectural scale models, mock-ups show the intent, the role and the relationships among the components. They help to focus the discussion on operational, rather than conceptual issues. The mock-up will also help us convey the ideas embodied in this research to current and potential sponsors, and help members of the research team to better understand their individual roles in the overall project. This task will develop a

PowerPoint or Flash presentation of the envisioned system, and will be revised throughout the project to reflect changes. The initial emphasize will be on the configuration of the architectural design components subject to a set of feasibility and cost constraints. Incorporating simple collaboration and negotiation constructs will follow.

Task 1: Develop the means to capture and represent the design context (the room)

This task addresses the 'custom' part of the project: modeling the environment into which the objects procured through the system will be fitted. We will develop (or adapt) the tools needed to build a simplified 3D model of the space, and the means to super-impose upon it digitataken off the customer's actual room or office. Only one room at a time will be modeled (unless and until we find it necessary to model whole suites). The tools needed for simple 3D modeling will be developed through VRML or Java applications (in fact, the envelop of the room needs to be only paper-thin, thus eliminating the overhead associated with transmitting volumetric data over the Web). In addition to walls, windows, doors, floor, ceiling, and other features of the space, objects of significance will also be modeled (e.g., fireplaces, built-in bookcases, etc.). We will develop a library of generic, parametric 'container' objects, of different geometries and topologies, so they can be picked from a template and adapted to model the actual room. Along with these modeling tools, we will also developed the step-by-step protocol that will guide users through the capture and modeling of their spaces (in a manner that will be similar to methods currently used by home & garden design centers, e.g., using gridded paper with cut-out furniture).

Task 2: Develop the means to capture and represent content (the furniture)

This task addresses the modeling and database aspects of the objects that can be placed in the room, including objects already owned by the customer and those procured from the seller, and their associated business attributes. We will develop compact 3D representations of furniture and other items, along with the data needed to make informed judgments about their procurement (including prices, delivery times, etc.). The 3D modeling part is a tedious but well-known effort that can be handled through a variety of 3D modeling software, such as 3D Studio Max. The output of these models will be VRML or Java3D applets.

The database associated with these objects will include not only *factual* information related to the objects, but also *operational* information. For example, appliances such as ovens and microwaves require certain electrical hookups, clearances (e.g. the required free space between a cook top and the upper cabinets of a kitchen), and other environmental considerations (ventilation, lighting, etc.). Such data is needed to make informed decisions about the procurement of these objects.

Object data models have been developed by many suppliers and researchers (including us). The most recent development comes in the form of IFCs (Industry Foundation Classes), developed by the IAI (International Alliance for Interoperability) for building components. It is not our intent to duplicate such standards, rather use them as an example for developing the database needed for the purposes of this project. On the other hand, office supplies and furnishing data is increasingly represented in e-catalogs using *XML* . Since the e-catalogs products typically include powerful search features, we will investigate the possibility of using such products if they support the representation and manipulation of our design projects or they can interface effectively with the design objects database.

Task 3: Develop e-commerce negotiations, collaboration, and coordination tools

This task will develop business components to support negotiation and collaboration processes associated with the configuration of a solution. The business coordination support is needed in various contexts. In the context of negotiation significant coordination is necessary for scenarios that involve configuring a solution from products and services purchased from ***S*** and ***G2*** (see Figure 7). In the context of negotiation, coordination is needed in order to adhere to the agreed

upon protocol. Designing efficient and practical negotiation and coordination protocols is essential to the business feasibility of the configurator. While other functions require coordination, the emphasize of this project is on the above two contexts. This task will address the back-end processes needed to manage ordering, billing, inventory, etc. in a simplified way, and will identify the hookup points or use existing software.

Accomplishing this task successfully requires the gathering of functional business requirements, developing a conceptual framework, and specific protocols. We plan to obtain the functional requirements through case studies and surveys. The particular negotiation methods will be based on the functional requirements and our previous work in this area (multi-attribute bidding techniques is likely to be applicable to the context of this project).

Task 4: User interface and communication protocol

The user interface is a kind of communication manager that will provide the necessary functionality to connect the different components of the system into one unit. Several interesting research questions will be addressed, including the "punch-out" capability interface and the navigation schemes supported by the configurator. "Punch out" (a term that was introduced by Ariba in the context of their eProcurement product) refers to the ability of the user to leave the primary purchasing system environment (e.g., the local or hosted e-catalog) and interact directly with a supplier. In the proposed project we will extend the concept to negotiations and architectural design where the "punch-out window" can be used to move from one context of negotiations (e.g. auction) to another (e.g., 1-on-1 bargaining), look for complimentary products or substitutes, or obtain expert advice for particular functions (see also next task).

Task 5: Develop on-line consultation

Choosing furniture and arranging it in an effective, economical, and aesthetically pleasing manner is a complex task. It often requires expert advice, provided by a design professional such as an architect, an interior designer, or a trained sales associate. We will not attempt to develop an artificial design 'agent,' thereby avoid the difficulties associated with building AI-like entities. Instead, we will provide the means for a real 'live' person to log-on, review the design concurrently with the customer or a-synchronously, and discuss with her options and opportunities. This on-line chat may use an avatar (we will decide that feature later). It will certainly include the possibility to mark-up the scene, and have both real-time and a-synchronous voice or text exchange. We envision the customer choosing from a list of consultants (who will make their services available for a fee), and sign up for a consultation 'session.'

Task 6: Interoperability

This task addresses issues related to the technology architecture of the proposed configurator to support distributed implementation. While this is not a major part of the proposed research, it is important to evaluate the technical architecture and business process protocols of the configurator relative to both business practices and technical standards. Integration with legacy systems as well as business-to-business system interoperability (e.g., in the context of "punch-outs") are crucial in order to facilitate commercialization of the prototype down the line. We expect to draw on other projects in the area of Enabling eBusiness, which are pursued by CITM under the direction of Professor Segev.

Task 7: Validation

The final task involves the testing, verification, and adjustment of the prototype. The testing will be done first with students, then with focus groups.

Tasks Implementation

The tasks will be implemented using the resources listed in the budget and follow the time line presented in the schedule section. In addition to the PIs, the students and post-docs, LynuxLine developers and the consultant will play a major role in the development and implementation tasks. We have identified Ms. Linda Haskin (or an equally qualified individual) as the type of consultant who is crucial to the success of this project. While a formal hiring process will be used to hire the consultant, Ms. Haskin will play a major role in this project to coordinate the activities of the PIs with those of LynuxLine, and provide advise on the commercial applicability of the various project components.

Schedule

	Year 1	Year 2	Year 3
Task 0: Mock-up	This task will be completed in the first three month of the project, with periodic updates throughout the project to reflect changes.		
Task 1: Capture and representation of the context	Develop parametric, template-based room geometry/topology, with the ability to apply texture-mapped digital images taken from the customer's own room.	Develop interactive modeling tool to let customer input her own room, and texture-map it with digital images.	Refinement of the same.
Task 2: Capture and representation of the content	Develop small number of template-based objects, with simple database. Investigate the functionality of e-Catalogs as related to the needs of this project.	Increased number of template-based objects, with the ability to modify textures, colors, and properties. Design and implement initial version of catalog content-management	Develop modeling tool to let customer build her own objects, texture map them, and assign properties. Develop advanced e-Catalog design and integrate the various content- related functionality.
Task 3: Negotiations, collaboration, and coordination methods and tools	Conduct case studies and surveys on functional requirements. Assess various negotiation and collaboration techniques as applicable to this project. Develop an initial conceptual framework for coordination.	Develop specific negotiation protocols for the scenarios addressed by this project. Design the collaboration framework. Implement initial version of negotiation, collaboration, and coordination protocols.	Develop and implement advanced versions of the negotiation, collaboration, and coordination protocols.
Task 4: User interface and communication protocol	First version of user interface and communication protocol. Investigate "punch-out" schemes.	Second version of user interface and communication protocol. First version of integrated configurator's interface and "punch-out" for negotiation, collaboration and expert advice.	Final version of user interface and communication protocol. Final version of integrated configurator's interface and "punch-out" for negotiation, collaboration and expert advice.
Task 5: On-line consultation	Shared viewability of layout with text-based, annotated suggestions.	Interactive, real-time chat with markup capabilities.	Avatar-based, interactive, voice enabled real-time interaction.
Task 6: Interoperability	This task will be performed throughout the project, ensuring that the technical architecture and business processes of the configurator are consistent with marketplace developments and business reality.		
Task 7: Validation	This task will be carried out at the end of each project year, to test and validate the research direction and results to that date.		

Deliverables	Overall design of the system, with basic prototype suitable for concept validation.	First operational prototype, suitable for field testing.	Complete prototype, suitable for technology transfer.

References

Canter D. (1977). *The psychology of place*. St. Martin press, NY.

Harrison S. and P. Dourish (1996). "Re-place-ing space: the roles of place and space in collaborative systems." Proceedings of CSCW'96 (an ACM publication).

http://moo.arch.usyd.edu.au:7778/vcguide2.pdf

http://www.eonreality.com

http://www.guggenheim.org/exhibitions/virtual/virtual_museum.html

http://www.activeworlds.com/

http://www.blaxxun.com/

Kalay Y.E. and J. Marx (2001). "Architecture and the Internet: designing places in cyberspace." *ACADIA'01* (W. Jabbi, ed.), Buffalo, NY.

Kalay Y.E. and C.H. Séquin (1995). "Designer-Client Collaboration in Architectural and Software Design." *ACADIA'95* (B. Kolarevic, ed.), Seattle, Washington.

Kalay Y.E., L. Khemlani and J.W. Choi (1998). "An Integrated Model to Support Distributed Collaborative Design of Buildings," *Automation in Construction*, **7**(2-3):177-188.

Kalay Y.E. (1998). "P3: An Integrated Environment to Support Design Collaboration," *Automation in Construction* **8**(1):37-48.

Kalay Y.E. (1999). "The Future of CAAD: from computer-aided design to computer-aided collaboration." *CAAD Futures'99* (Augenbroe & Eastman, eds.), Kluer Academic Publishers, NY.

Kalay Y.E. (1999). "Enhancing Multi-Disciplinary Collaboration Through Semantically-Rich Representation." *4th Design Methods Research Conference* (Porter & Goldschmidt, eds.), Cambridge, MA.

Kalay Y.E. and G. Black (2000). "Berkeley's A/E/C Collaborative Design Studio." *ASCE ICCCBE-VIII 8th International Conference on Computing in Civil and Building Engineering*, Stanford University.

Khemlani L., A. Timerman, B. Benne and Y.E. Kalay (1998). "Intelligent Representation for Computer-Aided Building Design," *Automation in Construction* **8**(1):177-188.

Novak M. (1992). *"Liquid Architectures in Cyberspace*." *Cyberspace: First Steps* (M. Benedikt, Ed.), pp. 225-254, MIT Press, Cambridge, MA.

Peri C. and Y.E. Kalay (2000). "ARCHVILLE: A distributed VR system for teaching real-time design collaboration." *ACSA Technology Conference*, MIT.

Sherry J.F. (1997). *Servicescapes: The Concept of Place in Contemporary Markets*. NTC Business Books, Lincolnwood, IL.

www.diarioelpais.com/muva2/

Appendix

APPENDIX: Twenty (20) collated sets of all appendix material must be submitted through the Contracts and Grants Office with other required forms so as to be postmarked no later than the proposal deadline. Identify each item with the name of the Principal Investigator. Do not use the appendix to circumvent the page limitations of the research plan. An application that does not observe these limitations may be returned.

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